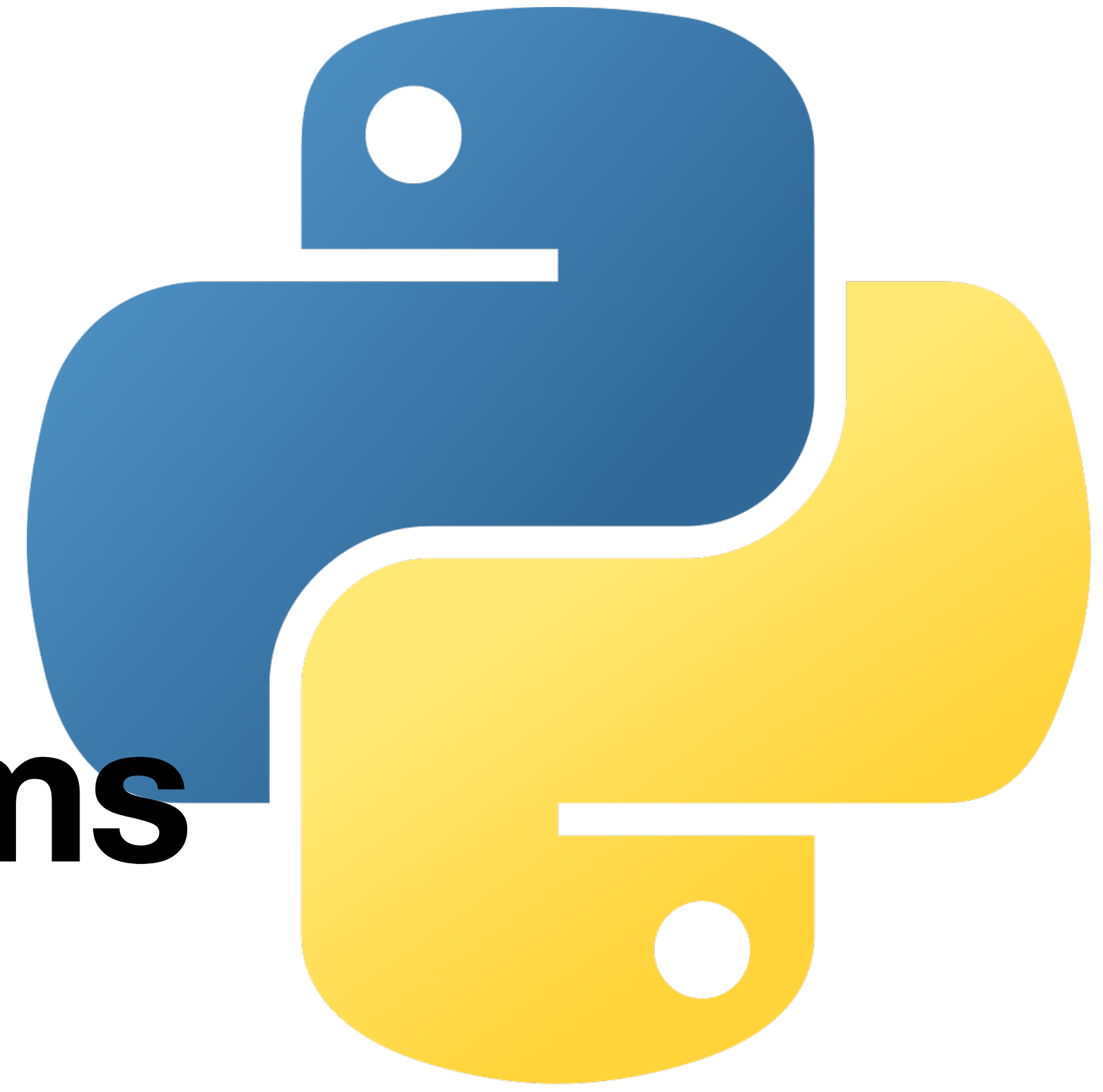


Root Finding Algorithms

C01 - Winter 2026



Tuesday, February 10th, 2026

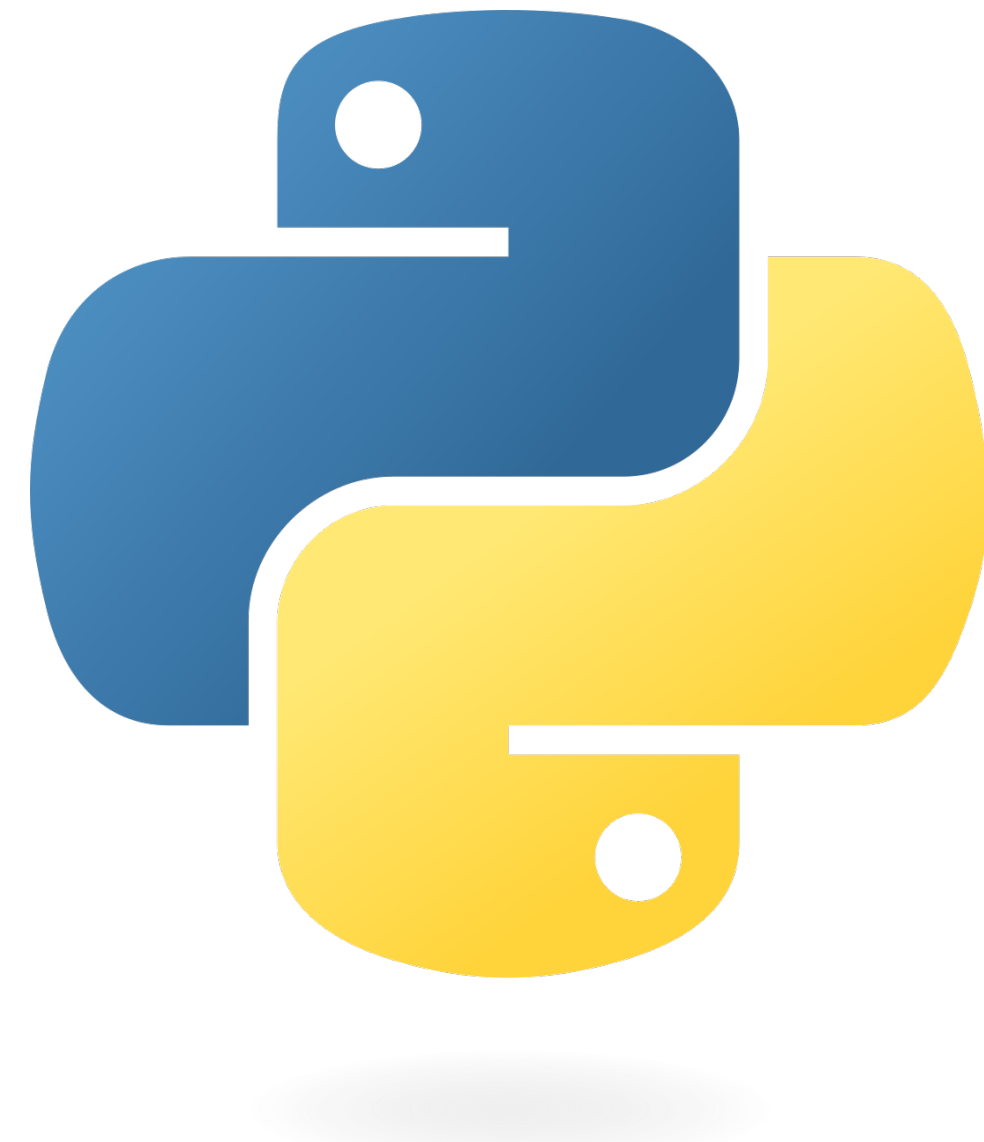
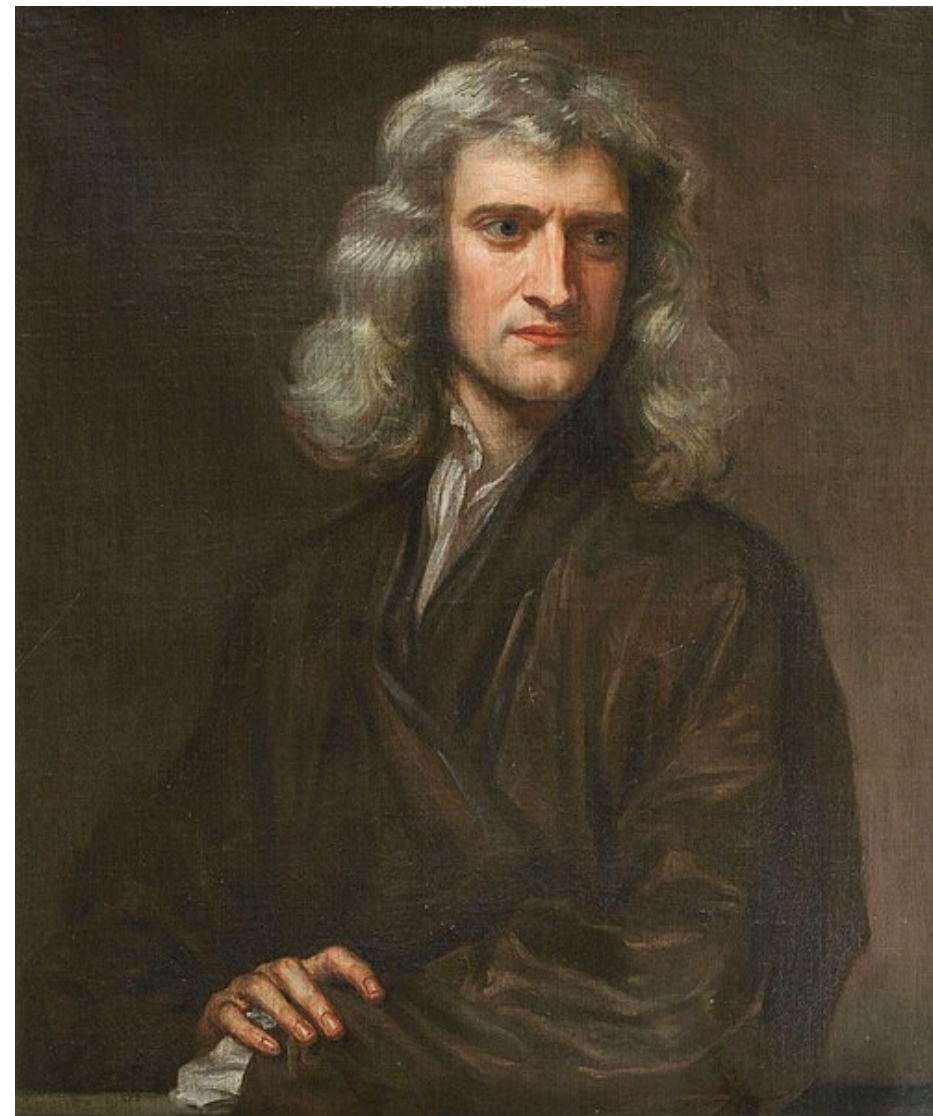
Last time

- **Question:** Given a function $f : \mathbb{R} \rightarrow \mathbb{R}$, how can we determine values $x \in \mathbb{R}$ (if they exist), such that $f(x) = 0$?
- Last time we saw that if f is continuous and changes sign on the interval $[a, b]$, then IVT allows us to do **Bisection Method**!
- However, this had some shortcomings...
- What are some disadvantages of bisection method?



Introducing Newton's Method

- Another very popular root finding method is **Newton's method**. It gains the strength of faster convergence and the ability to detect even multiplicity roots, but has its own drawbacks (we will discuss these as they come up).



Newton's method assumptions

In order to apply bisection method we needed the following assumptions: $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and changes sign exactly once on $[a, b]$.

In order to apply Newton's method we need the following assumptions: $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous on $[a, b]$, f' exists everywhere on $[a, b]$, and f has a root α somewhere inside of $[a, b]$.

Note: Speed of convergence will depend heavily on whether or not $f'(\alpha) = 0$.

Newton's method - 1

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function with derivative on $[a, b]$ and one root $\alpha \in [a, b]$. Begin with an initial guess x_0 for where the root occurs. Let

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

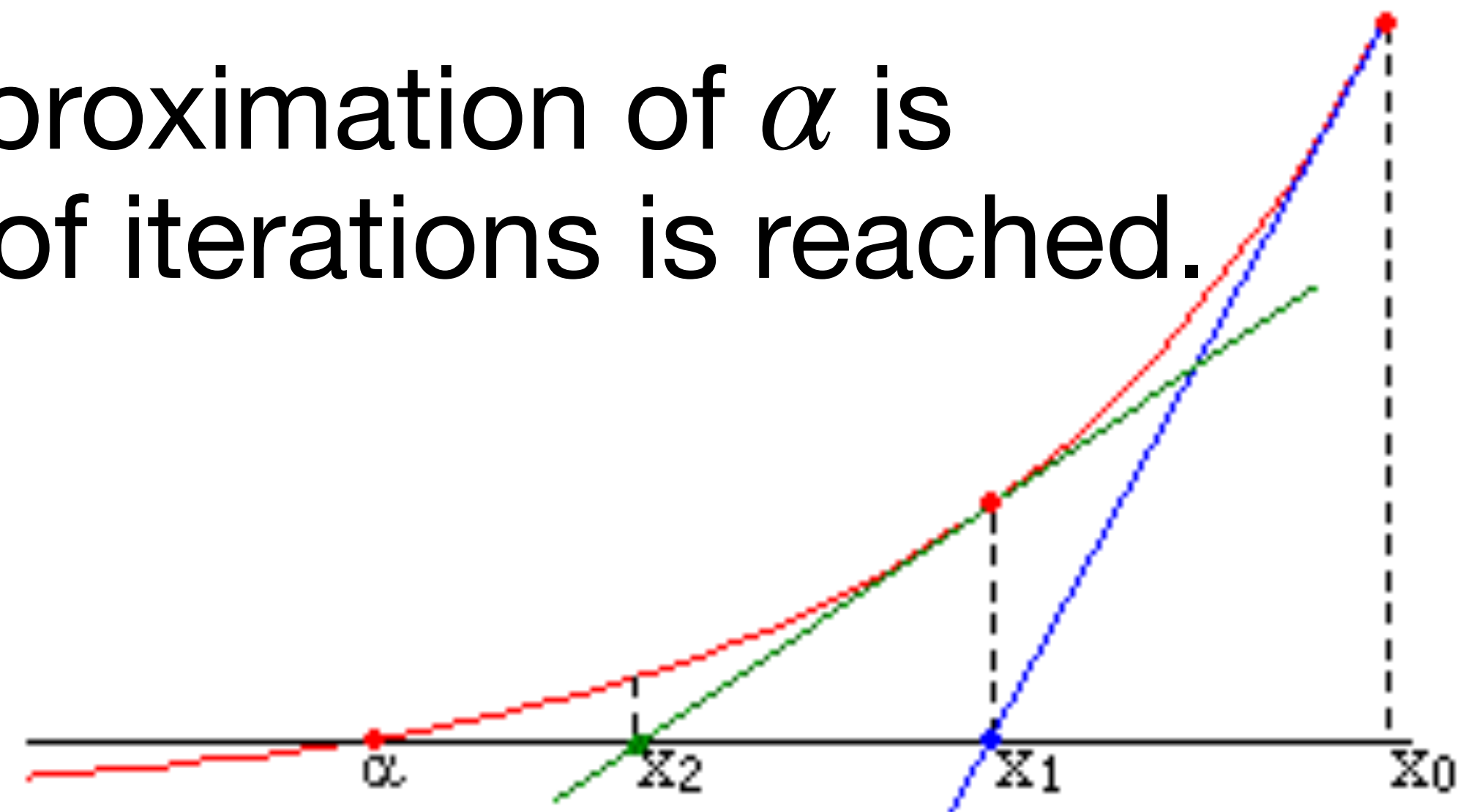
Notice: Geometrically, x_1 is exactly the root of the tangent line to f at the point x_0 .

Newton's method -2

Repeat this process as follows:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Stopping when an sufficiently good approximation of α is obtained or when a maximum number of iterations is reached.



Contrasting Newton and Bisection

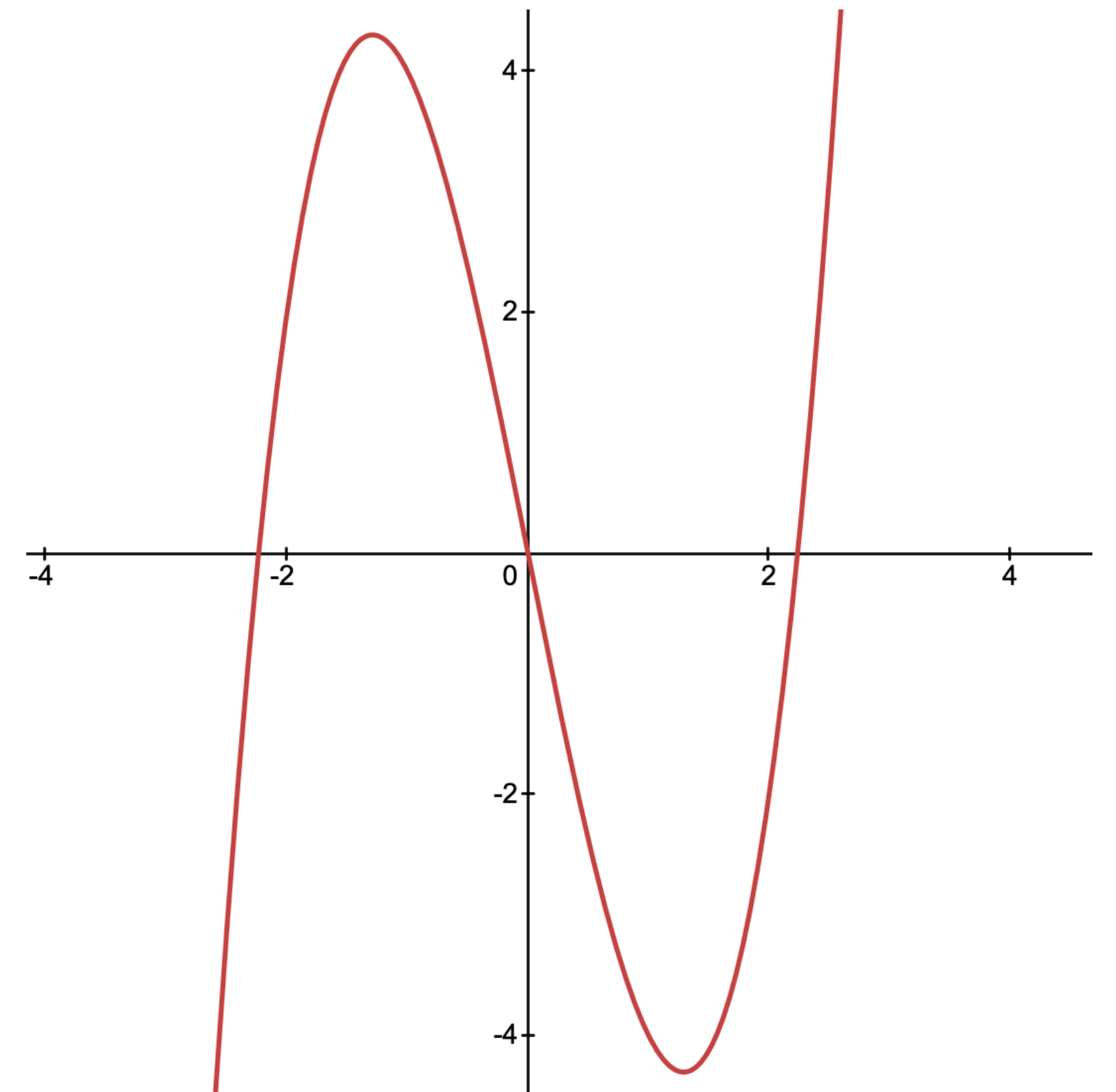
Newton's method comes with an extra requirement that bisection does not ask for. Namely, the existence of a way to compute derivatives. As this is a relatively big ask, Newton's method should give us some benefits for this. Some benefits of Newton's method are:

- Faster convergence. Bisection converges in linear time, that is each step reduces the length of the interval (and as such the error) by a factor of one half. Newton's method "can" converge quadratically.
- Newton's method can find roots of even multiplicity.

Contrasting Newton and Bisection

Newton's method does however have some key drawbacks. Namely, computing derivatives accurately can be computationally difficult and Newton's method can fail to converge to a root.

Example: $f(x) = x^3 - 5x$ with
initial guess $x_0 = 1$.



Newton's method failure

Given $f(x) = x^3 - 5x$ with initial guess $x_0 = 1$. We can see that

$$x_1 = x_0 - \frac{(1)^3 - 5(1)}{3(1)^2 - 5} = 1 - 2 = -1$$

$$x_2 = x_1 - \frac{(-1)^3 - 5(-1)}{3(-1)^2 - 5} = -1 + 2 = 1$$

This will oscillate endlessly.

Now go code up Newton's Method

A variant of Newton's method

A main drawback of Newton's method is that we need to know how to compute the derivative of f .

Later in this course, we will discuss at length how to compute the derivatives of functions (at least numerically). However, there does however exist a variant of Newton's method that does not require us to know how to compute the derivative of f .

Secant method - 1

Secant method works identically to Newton's method, except we replace the derivative of f at a point, with the slope of a closely related secant line about this point. That is, here

$$x_{n+1} = \frac{x_{n-1}f(x_n) - x_nf(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

Note: In order to approximate a root with the secant method, we need to provide two initial guesses (in order to compute the secant line).

Secant method - 2

The formula in the previous slide for the n -th approximation of the secant method comes from the formula for the secant line. Recall that the equation of the secant line to f at points $a \leq b$ is given by:

$$y = \frac{f(b) - f(a)}{b - a}(x - b) + f(b)$$

Thus, use the secant line as an approximation of the tangent line to f at a point $c \in [a, b]$ (this is a reasonable approximation if $[a, b]$ is small!)